Syed Abdul Basit

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Research Interests

LLMs as verification agents for continual-learning systems; self-improving and multi-agent LLM systems; LLM-as-judge evaluation and calibration against human labels; clinical AI.

Education

M.Sc. in Artificial Intelligence Engineering, Thesis Track Jul. 2025 – Expected 2027
Okan University (English-medium) Istanbul, Turkey

- GPA 3.43/4.00. Coursework (AA): Machine Learning, Deep Learning, Design of AI Systems, Introduction to AI Engineering.
- Thesis (approved): "Using LLMs as Verification Agents for Continuous Learning Systems." Supervisor: Sina Alp.

B.E. in Mechatronics Engineering Aug. 2014 – Feb. 2020
Shaheed Zulfikar Ali Bhutto Institute of Science and Technology (SZABIST) Karachi, Pakistan

- Best Innovative Design Award.

Publications

Peer-reviewed conference papers *(* = oral presentation)*

[C2] **Syed Abdul Basit**, Ahmed Anas. "PhyS2S-GNN: A Physics-Informed Dynamic Graph Neural Network for Compound Extreme Weather Event Prediction at Sub-Seasonal Timescales." *IEEE Innovations in Intelligent Systems and Applications Conference (ASYU)*, 2026. Accepted, to appear in IEEE Xplore.*

[C1] **Syed Abdul Basit**. "Bidirectional Cross-Modal Attention for Multimodal Disease Prediction: Fusing Clinical Narratives with Vital Signs." *IEEE Innovations in Intelligent Systems and Applications Conference (ASYU)*, 2026. Accepted, to appear in IEEE Xplore.*

Research Experience

Graduate Researcher, Physics-Informed Graph Learning 2026
Okan University; supervisor: Prof. Mahsa Mikaeili Istanbul, Turkey

- 14-day prediction of compound heatwave–drought events. Designed PhyS2S-GNN (dual-LSTM encoder, dynamic graph attention over a 48-node graph, physics-informed loss, dual decoder) and compared it with LSTM, GCN and GAT baselines [C2].

Graduate Researcher, Multimodal Clinical Prediction 2026
Okan University; supervisor: Sina Alp Istanbul, Turkey

- Disease prediction from symptom narratives and vital signs. Designed a bidirectional cross-modal attention model (BioClinicalBERT, TabNet-style encoder) and compared it with single-modality, concatenation and unidirectional baselines [C1].

Applied ML Research in Industry

Senior Machine Learning Engineer Mar. 2025 – Present
Applied AI Istanbul, Turkey

- *LLM multi-agent system.* Designed a three-agent LangGraph pipeline (Extractor → Validator → Router) with a dedicated validation stage between agents and escalation to human reviewers; runs on 10K+ documents per month.
- *Evaluating LLM outputs.* Built an evaluation protocol combining RAGAS metrics, LLM-as-judge scoring and a 200-item golden set, run as a regression test on every release; detected 3 critical regressions across 14 releases.
- *LLM feature extraction.* Used an LLM to turn unstructured complaint text into structured features for a gradient-boosted classifier (4 classes, 12K labelled records): AUC-ROC 0.94, macro-F1 0.89 on a held-out 20% split.

ML & Data Scientist Sep. 2020 – Mar. 2025
Aga Khan University Hospital (AKUH) Karachi, Pakistan

- *Clinical text classification.* Diagnosed why a first triage classifier reached only 71% F1 (class imbalance, domain mismatch), then rebuilt it with balanced sampling, domain-specific tokenisation and LoRA fine-tuning: 91% macro-F1 on 8K held-out records vs. 67% for zero-shot GPT-3.5. Deployed with the clinical team.
- *Clinical data quality.* Designed automated schema validation and drift detection for 50K+ daily records across a multi-country hospital network (Spark, Azure Data Factory), reducing downstream error rate from 40% to under 4%.
- Supervised a team of 4 and mentored junior data scientists.

Open Source and Projects

- *PyHealth (open-source healthcare AI library, UIUC).* Pull request #1324: fixed the shape check and documentation of `regression_metrics_fn` and added 10 unit tests (2026).
- *Token-efficient self-improving coding agent* (github.com/absyed-lab). LangGraph loop of coding, critic and LLM-as-judge agents; the judge decides when further self-improvement attempts are no longer worth the tokens. Work in progress.

Skills

Machine learning	PyTorch, PyTorch Geometric, Transformers, LoRA fine-tuning, scikit-learn, XGBoost
LLM systems	LangGraph multi-agent pipelines, LLM-as-judge evaluation, RAGAS
Engineering	Python, SQL, Spark, Docker, FastAPI, GitHub Actions, AWS, Azure, Git
Certifications	Claude Certified Architect – Foundations; AI Fluency (Anthropic, 2025)
Languages	English (C2), German (C1), Urdu (native), Turkish (conversational)